HARIDEV P

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Summary

Offensive-security focused Computer Science student with hands-on experience in penetration testing, Active Directory exploitation, and red team operations. Passionate about adversary simulation, C2 infrastructure development, post-exploitation tradecraft, and building security tooling across network and infrastructure attack surfaces.

Experience

Security Research Intern **May 2026 – Present**
ASTRA Lab (CyStar), IIT Madras *Chennai, Tamil Nadu*

- Conducted offensive security research under Prof. Narayanaswamy N S and Dr. Jaimandeep Singh (Senior Consultant, ex-MoD); contributed to **CipherC2**, a confidential lab project modelling adversary C2 TTPs and **initial access methodologies** aligned with the **MITRE ATT&CK framework**.
- Applied Windows internals knowledge to research evasion strategies and **initial access vectors** for adversary emulation scenarios.

Projects

Stealth C2 Framework | GitHub | v1.0 Release | *Python, Discord API, AES-256-GCM* **Nov 2025 – Jan 2026**

- Built a modular red team C2 framework tunnelling AES-256-GCM encrypted traffic over the Discord API to blend with legitimate HTTPS traffic and bypass firewall egress restrictions.
- Implemented **emoji-based steganographic command protocol** mapping 7 operator commands (RCE, exfiltration, surveillance, persistence) to innocuous chat messages; integrated anti-forensic detection across 5 analysis tools (Wireshark, ProcMon, Cuckoo) with automated kill switch.

Enterprise Active Directory Attack Lab | *Windows Server 2022, PowerShell, Impacket* **Nov 2025 – Dec 2025**

- Built a 4-machine virtual enterprise environment (Windows Server 2022 DC, 2 Windows 11 workstations, Kali attacker) simulating a real-world corporate network for adversary simulation.
- Executed a full red team attack chain – LLMNR poisoning, Kerberoasting, SMB relay, Pass-the-Hash, and privilege escalation – achieving Domain Admin access while bypassing Windows Defender default configurations.

AI Automated Pentesting Framework | *FastAPI, Celery, ChromaDB, LangChain, PostgreSQL* **Jan 2026 – Present**

- Building a RAG-based pentesting assistant (team of 4) that retrieves contextual CVE data and attack patterns to generate adaptive, behaviour-driven engagement plans against live targets.
- Implementing LLM-driven behavioural analysis of target system responses to dynamically adjust attack strategies mid-engagement; targeting conference publication 2026.

Talks

Initial Access Vectors | *DeepTrustxAI Offensive Security Workshop, Session 2*

- Presented a session on **Initial Access Vectors (IAV)** and adversary tradecraft to an audience of industry security professionals and law enforcement officials.

Publications

Lane Change Prediction & Obstacle Avoidance in Autonomous Vehicles Using Rainbow DQN

- Haridev P (corresponding-author) – "An Adaptive Framework for Lane Change Prediction and Obstacle Avoidance in Autonomous Vehicles Using Rainbow DQN," *SCRS CIMA 2026*, NIT Puducherry (Springer LNNS, Scopus-indexed) – *Accepted & Presented.*

Technical Skills

Languages: Python, C++, SQL, HTML/CSS, JavaScript, Bash
Security Tools: Nmap, Burp Suite, Wireshark, Impacket, John the Ripper, Metasploit, BloodHound
Systems: Linux (Kali, Fedora), Windows Server, Active Directory
Concepts: Penetration Testing, Red Teaming, C2 Infrastructure, Post-Exploitation, Privilege Escalation, Network Enumeration, AD Exploitation, Malware Analysis
Developer Tools: Git, Docker
Platforms: TryHackMe (ArtemizZ – Top 30%), Hack The Box (Level 12 – 13 machines, Guided Tracks Tier 0–2)

Education

Amrita Vishwa Vidyapeetham **Coimbatore, Tamil Nadu**
B.Tech in Computer Science and Engineering (CGPA: 7.1) *Aug 2023 – Jul 2027*
CKG Memorial HSS, Chingapuram **Kozhikode, Kerala**
XII – 96% *Jun 2021 – Mar 2023*

Certifications

Google Cybersecurity Certificate | *Coursera* **[Credential] | Nov 2024**

TryHackMe – Advent of Cyber | *TryHackMe* **[Credential] | Dec 2025**